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Threat, Vulnerability, Risk and Exploit

1. Definitions

Term	Meaning	Example
Threat	A person, event or condition capable of causing harm.	A criminal attempts to steal a company's customer data.
Vulnerability	A weakness that could be exploited.	A server runs outdated software containing a known security flaw.
Risk	The possibility and impact of harm when a threat exploits a vulnerability.	The company may suffer data loss if the exposed server is compromised.
Exploit	A method, code or technique used to take advantage of a vulnerability.	A proof-of-concept demonstrates the impact of a known flaw in an authorized test environment.

2. How They Relate

A threat may exploit a vulnerability using an exploit, potentially causing an incident. Risk depends on factors such as likelihood, exposure, existing controls and potential impact.

A vulnerability does not automatically mean that a system has been compromised. Similarly, an exploit may exist without a successful attack.

3. Example Scenario

A company uses an internet-facing application that has not been patched.

- Threat: An attacker seeks access to company records.
- Vulnerability: The application has a known, unpatched flaw.
- Exploit: A technique could take advantage of that flaw.
- Risk: Confidential records may be accessed or the service may be disrupted.

4. Risk Reduction

- Patch vulnerable software.
- Limit unnecessary network exposure.
- Apply least-privilege access controls.

- Monitor logs and suspicious activity.
- Maintain backups and incident response procedures.
- Prioritize remediation based on likelihood and impact.

5. Conclusion

These terms describe different parts of security analysis. Distinguishing them helps defenders assess weaknesses, prioritize fixes and explain potential consequences to decision-makers.

